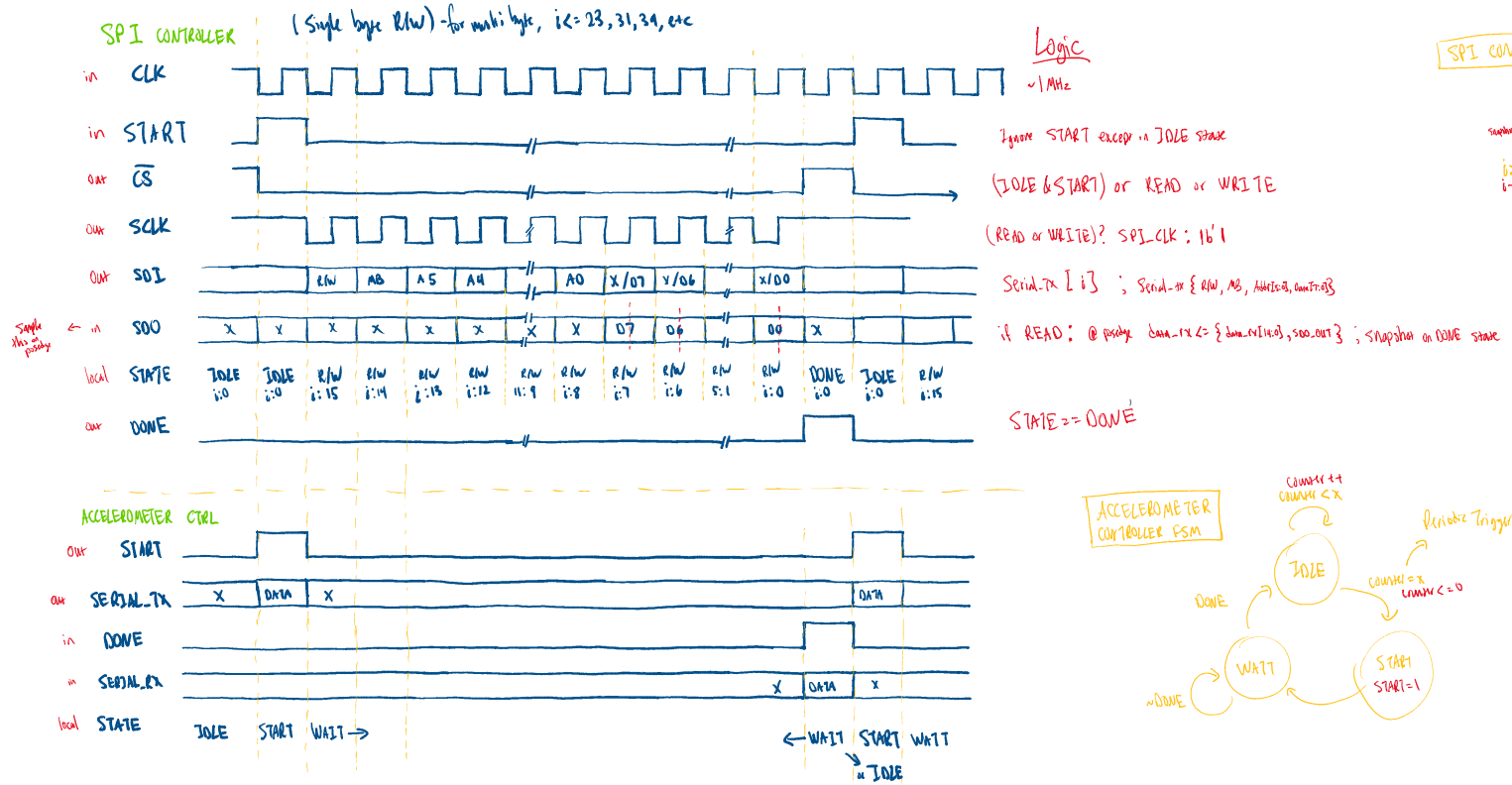
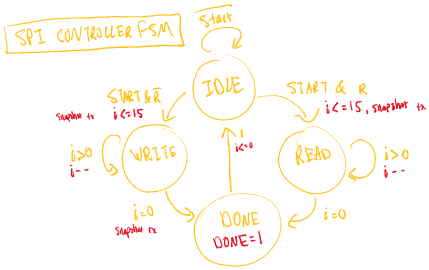
FSM Design, Timing Diagrams & Testbench Results

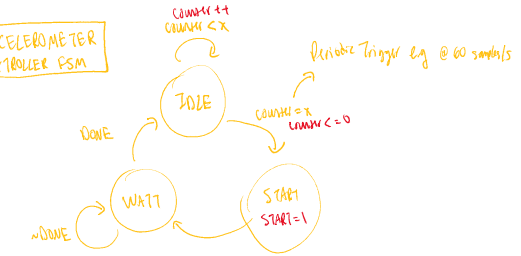
Monday, November 2, 2026 10:12 PM

Rising Edge
→ Sample SDO

falling edge
→ Update States
→ update signals



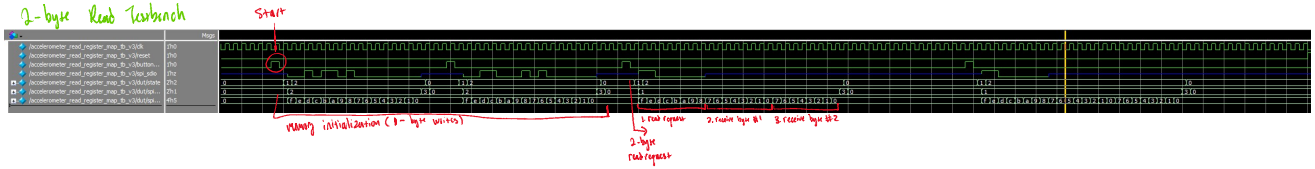




Accelerometer Memory Initialization Testbench



2-byte Read Testbench



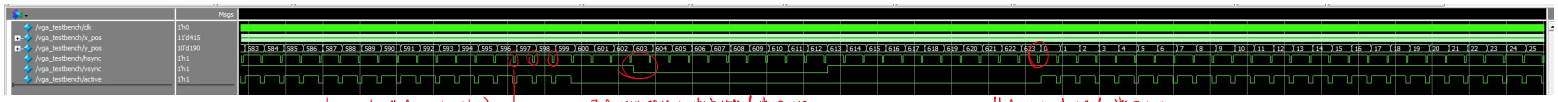
Device IO Read request (on-board verification, not a simulation)



Read 0x00 request

received byte; 0xE5 as expected

VGA Signal Generation Testbench

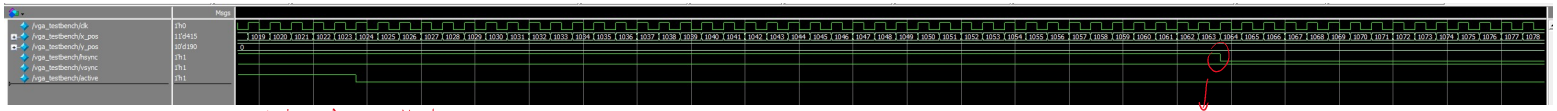


1. y_pos counts the scanline # (vertical position)

2. Deassert hsync once at the end of each scanline

3. Deassert vsync at the bottom of the screen

4. Return to the top of the screen.



Zoom-in: 1. x_pos counts the horizontal position within the scanline;

2. deassert hsync at the end of the scanline